**Mohammad Zhalechian**

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## APPOINTMENT

**Indiana University, Bloomington, IN** *2023 – Present*
Assistant Professor of Operations and Decision Technologies
Kelley School of Business

**Harvard University, Cambridge, MA** *2022 – 2023*
Postdoctoral Fellow
Harvard Kennedy School

## EDUCATION

**University of Michigan, Ann Arbor, MI**
Ph.D. in Industrial and Operations Engineering (Operations Research) *2022*
M.A. in Statistics *2019*

**University of Tehran, Tehran, Iran**
M.Sc. in Industrial Engineering and Operations Research *2017*

**K. N. Toosi University of Technology (KNTU), Tehran, Iran**
B.Sc. in Industrial Engineering and Operations Research *2014*

## RESEARCH INTERESTS

- Sequential and Data-Driven Decision-Making
- Design and Evaluation of Human-AI Systems
- Algorithm Applications in Healthcare and High-Stakes Settings

## PUBLICATIONS

[† industry collaborators; (M.D.) M.D. collaborators]

### Published Papers

[1] **Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy**
***Management Science, 2026***
M. Zhalechian, S. Saghafian, O. Robles†

- *Media coverage (Harvard Kennedy School):* A Human-Algorithm Approach to Approving Medical Devices Could Slash the Recall Rate and Save Billions
- *Media coverage (INFORMS press release):* AI Could Cut Medical Device Recalls by Nearly a Third
- *Media coverage (Medical Xpress):* AI Could Cut Medical Device Recalls by Nearly a Third
- *Media coverage (Today's Medical Developments):* Artificial Intelligence Could Reduce Medical Device Recalls
- *Media coverage (Devdiscourse):* Transforming FDA Clearance: How AI and Human Insight Can Improve Medical Device Safety

[2] **Contextual Learning with Online Convex Optimization with Applications to Medical Decision-Making**
***Management Science, 2025***
Collaboration with Massachusetts Hospital
E. Keyvanshokooh, M. Zhalechian, C. Shi, MP. Van Oyen, P. Kazemian

- Finalist, INFORMS Minority Issues Forum (MIF) Paper Competition, 2026
- Finalist, POMS College of Healthcare Operations (CHOM) Best Paper Competition, 2022
- Finalist, INFORMS MSOM Best Student Paper Competition, 2021
- Finalist, INFORMS Health Applications Society (HAS) Best Student Paper Competition, 2021
- Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper, 2020

[3] **Dynamic Care Unit Placements under Unknown Demand with Learning**
***Manufacturing & Service Operations Management, 2025***
Collaboration with St. Joseph Hospital
A. Dean, M. Zhalechian, MP. Van Oyen

[4] **Data-driven Hospital Admission Control: A Learning Approach**
***Operations Research, 2023***
Collaboration with St. Joseph Hospital
M. Zhalechian, E. Keyvanshokooh, C. Shi, MP. Van Oyen

- Winner, IOE Richard C. Wilson Best Student Paper Award on Service Systems, 2021
- Finalist, Seth Bonder Scholarship, INFORMS Health Applications Society, 2020

[5] **Online Resource Allocation with Personalized Learning**
***Operations Research, 2022***
Collaboration with Michigan Medicine
M. Zhalechian, E. Keyvanshokooh, C. Shi, MP. Van Oyen

[6] **Validation of a Visual Field Prediction Tool for Glaucoma: A Multicenter Study Involving Patients with Glaucoma from Throughout the United Kingdom**
***American Journal of Ophthalmology, 2025***
Collaboration with Medisoft UK
A. Dean, DJ. Fu (M.D.), M. Zhalechian, MP. Van Oyen, M. Lavieri, AP. Khawaja (M.D.), J. Stein (M.D.)

[7] **Augmenting Kalman Filter Machine Learning Models with Data from Optical Coherence Tomography to Predict Future Visual Field Loss: The African Descent and Glaucoma Evaluation Study**
***Ophthalmology Science, 2022***
Collaboration with Kellogg Eye Center and Shiley Eye Institute
M. Zhalechian, MP. Van Oyen, M. Lavieri, C. Andrew†, J. Stein (M.D.), et al.

## Working Papers

[8] **From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement**
Under Review
Collaboration with Michigan Medicine
A. Dean, M. Zhalechian, I. Attari, M. Garifullin†, J. Pardo†, MP. Van Oyen
- Implemented at Michigan Medicine since July 2023
- Finalist, INFORMS Service Science Cluster Best Paper Competition, 2026
- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026
- Accepted at Michigan Medicine Quality Month Symposium, 2024

[9] **Does Early Pain Management Pay Off? Evidence on Utilization, Costs, and Surgeries**
Under Review
Collaboration with Integrative Sports & Spine Pain Clinic in Southern California
B. Li, M. Zhalechian, C. Chen, N. Vaughn (M.D.), H. Lee (M.D.)

[10] **Learning Bundle Pricing of Reusable Resources**
Under Review
A. Dean, M. Zhalechian, C. Shi
- Finalist, Service Science Best Student Paper Competition, 2024

[11] **Operations Management for Restorative Public Health: Grand Challenges in Reversing the Damage of the Opioid Crisis**
Major Revision at ***Manufacturing & Service Operations Management***
C. Chen, J. Helm, B. Li, J. Mejia, M. Zhalechian (alphabetical authors)
- Selected for presentation at the Healthcare SIG Track, MSOM Conference, 2026

## Selected Published Papers Prior to Ph.D.

**Sustainable Design of a Closed-loop Location-routing-inventory Supply Chain Network Under Mixed Uncertainty**
*Transportation Research Part E*, 2016
M. Zhalechian, R. Tavakkoli-Moghaddam, B. Zahiri, M. Mohammadi

**Hub-and-spoke Network Design under Operational and Disruption Risks**
*Transportation Research Part E*, 2017
M. Zhalechian, SA. Torabi, M. Mohammadi

**A Self-adaptive Evolutionary Algorithm for a Fuzzy Multi-objective Hub Location Problem: An Integration of Responsiveness and Social Responsibility**
*Engineering Applications of Artificial Intelligence*, 2017
M. Zhalechian, R. Tavakkoli-Moghaddam, Y. Rahimi

## HONORS AND AWARDS

(∗ denotes honors or awards received by students)

**2026 Finalist, INFORMS Minority Issues Forum (MIF) Early Career Award**

**2026 Winner, INFORMS Junior Faculty Interest Group (JFIG) Teaching Excellence Award**

**2026 Winner, INFORMS Data Mining Society Teaching Award**

**2026 Finalist, INFORMS Service Science Cluster Best Paper Competition**

**2026 Finalist, INFORMS Minority Issues Forum (MIF) Paper Competition**

**2025 Operations and Decision Technologies Faculty Scholar for Research Excellence**

**2025 Kelley School of Business Research Award**

**2024* Finalist, Service Science Best Student Paper Competition**

**2022 Finalist, POMS College of Healthcare Operations (CHOM) Best Paper Competition**

**2022 Towner Prize for Distinguished Academic Achievement**, University of Michigan

**2022 Rackham Outstanding Graduate Student Instructor Award**, University of Michigan

**2021 Rackham Graduate Research Grant**, University of Michigan

**2021 Winner, IOE Richard C. Wilson Best Student Paper Award**, University of Michigan

**2020 Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper**

**2020 Finalist, INFORMS Seth Bonder Scholarship for Applied OR in Health Services**

**2020 IOE Brown Outstanding Graduate Student Instructor Award**, University of Michigan

**2018–2021 Rackham Travel Grant Awards**, University of Michigan

**2017 Graduate Student of the Year**, College of Engineering, University of Tehran

## INVITED SEMINARS AND CONFERENCE PRESENTATIONS

**From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement**

- MSOM Conference, Special Interests Group *Jul. 2026*

**Interpretable Design of Multi-Agent Mobile Health Applications**

- Center for the Business of Life Sciences, Kelley School of Business *Oct. 2026*
- Workshop on AI & Analytics for Social Good, College Park, MD *Apr. 2026*
- INFORMS Healthcare Conference, Toronto, Canada *Jul. 2023*

**Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy**

- Wayne State University, Industrial & Systems Engineering Seminar Series *Nov. 2025*
- INFORMS Annual Meeting (2023, 2024, 2025); MSOM Conference (2025); POMS Annual Conference (2024); INFORMS Healthcare Conference (2023)

**Data-Driven Learning and Resource Allocation in Service Operations Management**

- Dartmouth College, Tuck School of Business *Jan. 2023*
- Indiana University, Kelley School of Business *Jan. 2023*
- Carnegie Mellon University, Tepper School of Business *Dec. 2022*
- Rice University, Jones Graduate School of Business *Dec. 2022*
- University of North Carolina at Chapel Hill, Kenan-Flagler Business School *Dec. 2022*
- University of Illinois Urbana-Champaign, Gies College of Business *Dec. 2022*
- Miami University, Herbert Business School *Dec. 2022*

- University of Illinois Chicago, College of Business Administration *Dec. 2022*
- Baruch College, Zicklin School of Business *Dec. 2022*

**Personalized Hospital Admission Control: A Contextual Learning Approach**

- INFORMS Annual Meeting (2019, 2020, 2021, 2022); POMS Annual Conference (2019, 2021, 2022); INFORMS Healthcare Conference (2019)

**Online Advance Appointment Scheduling with Personalized Learning**

- INFORMS Annual Meeting (2020, 2021)

**Joint Learning and Optimization for Healthcare Operations and Medical Decision-Making**

- INFORMS Annual Meeting, Seth Bonder Scholarship Session *Nov. 2020*

**A Distributionally Robust Capacity Planning Model for Optimizing Access Delay in Surgical Services under Limited Information**

- INFORMS Healthcare Conference (2019); INFORMS Annual Meeting (2018)

## TEACHING EXPERIENCE

**Instructor, Indiana University, Bloomington, IN** *Spring 2024, 2025 & 2026*
Class: *Business Analytics and Modeling* (K353)
Three sections per year
Instructor rating: 6.5/7 average across nine sections; 6.6/7 most recent (program mean: 6.2)

**Instructor, University of Michigan, Ann Arbor, MI** *Fall 2020*
Class: *Economic Decision Making* (IOE 201)
Instructor rating: 4.9/5 (college median: 4.5)

**Recitation Instructor, University of Michigan, Ann Arbor, MI** *Winter 2020*
Class: *Introduction to Markov Processes* (IOE 316)

**Teaching Assistant, University of Michigan, Ann Arbor, MI**

- Practicum in Hospital Systems (IOE 481) *Fall 2021 – Winter 2022*
- Service Operations Management (IOE 419) *Winter 2021*
- Probability and Statistics for Eng. (IOE 265) *Fall 2018, Winter 2019, Fall 2019*

## ADVISING AND MENTORING

**PhD Student Committees**

- Ali Kaan Kurbanzade, Kelley School of Business, 2025–present
- Iman Attari, Kelley School of Business, 2023–2025
  Placement: University of Cincinnati (Lindner College of Business)

**Master's and Undergraduate Mentoring**

- Faculty Mentor for Master's Students, Faculty Assistance in Data Science (FADS) Program, Kelley School of Business
- Faculty Mentor for Undergraduate Students, Center of Excellence for Women & Technology, Indiana University

## ACADEMIC SERVICE

### Service to the Profession

- *Referee:* Management Science, Operations Research, Manufacturing & Service Operations Management, Production and Operations Management, Information Systems Research, European Journal of Operational Research, Omega, Health Care Management Science
- *Judge:* Public Sector OR Best Paper Competition, INFORMS, 2023

### Session Organizer & Chair for Conferences

**2026** INFORMS Annual Meeting (AI in Healthcare); INFORMS Healthcare Conference (AI for Healthcare Oversight and Operations & AI and Analytics for Decision-Making in Healthcare)

**2025** INFORMS Annual Meeting (Data-driven Healthcare Analytics)

**2024** POMS Annual Conference (Policies and Regulation in Healthcare)

**2021** INFORMS Annual Meeting (Sequential Decision Making and Machine Learning in Healthcare); POMS Annual Conference (Healthcare Operations Management)

**2020** INFORMS Annual Meeting (Predictive Analytics & Sequential Decision-Making in Healthcare)

### Department and School Service

- PhD Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2025
- Faculty Recruiting Committee, Operations and Decision Technologies, Kelley School of Business, 2024
- Faculty Recruiting Committee (Graduate Student Committee), Industrial and Operations Engineering, University of Michigan, 2021
- Chair, Anti-Racism and DEI Committee, INFORMS Student Chapter, University of Michigan, 2020–2021
- Chair, Professional Development Committee, INFORMS Student Chapter, University of Michigan, 2019–2020